

DID

Installation and operating instructions



Other languages

<http://net.grundfos.com/qr/i/99037650>

English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe Grundfos DID (Dosing Instrumentation Digital).

Sections 1-6 give the information necessary to be able to unpack, install and start up the product in a safe way.

Sections 7-13 give important information about the product, as well as information on service, fault finding and disposal of the product.

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1. General information

1.1 Target group

1.1.1 Qualification and training

The persons responsible for installation, startup and service must be appropriately qualified for these tasks.

If the persons do not have the necessary knowledge, training and instruction must be given. If necessary, training can be performed by the manufacturer or supplier on request.

1.1.2 Obligations of the operating company

- Observe the local safety regulations.
- Instruct the operating persons.
- Provide the stipulated safety equipment and personal protective equipment.
- Arrange regular maintenance.

1.1.3 Obligations of the user

- Read this manual thoroughly before taking the product into operation.
- Observe the recognised health and safety regulations as well as the accident prevention regulations.
- Wear protective equipment in accordance with national health and safety regulations when working on the system and handling chemicals.

1.2 Symbols used in this document



DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.



If these instructions are not observed, it may result in malfunction or damage to the equipment.

1.3 Symbols on the product

For symbols on the product see section [14. Appendix](#).



Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

2. Safety instructions

WARNING

Electric shock



- Switch off the power supply before connecting the power supply cable and relay contacts.
- Do not dismantle the control unit.
- Installation and connection of the device and the associated supplementary components must only be carried out by qualified persons.
- Maintenance and repair must only be carried out by qualified persons.
- Observe the local safety regulations.

WARNING

Toxic material



- Parts of the DID can be contaminated with dangerous chemicals or pathogenic germs.
- If the DID has been in contact with dangerous chemicals or pathogenic germs, wear protective clothing and take all necessary precautions to prevent endangering your health when installing or dismantling the DID.



Modification of the DID is strictly prohibited.

3. Receiving the product

3.1 Inspecting the product

- Check the received consignment for completeness.
- Check the DID for damage.
- Install as soon as possible after unpacking.
- Do not install or connect damaged products.

3.2 Transporting the product

- Before transporting the DID, empty all pipes, hoses and flow cells, remove the sensors and pack them separately.
- Observe the permissible ambient conditions. See section [12. Technical data](#).
- Transport the DID carefully, do not drop. Do not expose the DID to strong impact, mechanical loads or vibrations.
- Keep the DID away from corrosive substances, vapours of organic solvents, nuclear radiation and strong electromagnetic radiation.
- Use the original packaging or equivalent to protect the DID during transport.

4. Installation

WARNING

Electric shock



- Switch off the power supply before installation.
- IP65 only applies, if the housing cover and the terminal compartment of the CU 382 are properly closed, and the appropriate cable glands or dummy caps are fitted.
- Installation must be carried out by qualified and authorised persons.

4.1 Installation location

Correct installation is important for proper functioning of the DID.

- Make the DID easily accessible for all tasks.
- Provide sufficient space for all components of the DID.
- Avoid external interferences, such as electric and electro-magnetic interferences by leakage current, earth fault of pumps, electric motors, high voltage currents.
- Protect the DID from direct sunlight and rain.
- Provide a reliable power supply for the control unit.

See also section [12. Technical data](#) and the manual "Sensors V1".

4.2 Installation of the DID

DID with flow cell

All parts of the DID with flow cell are assembled on a mounting plate with pre-drilled holes for quick and easy installation.

- Use spacers (min. 10 mm) for installation of the DID with flow cell on a flat wall.

See also sections [8. Product introduction](#) and [12.4 Dimensions](#).

DID with sensor for tank immersion

The CU 382 control unit of the DID with sensor for tank immersion can be mounted directly on a wall. See also sections [8. Product introduction](#) and [12.4 Dimensions](#).

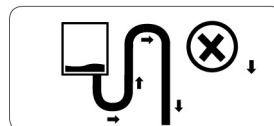
4.3 Hydraulic connection

See also the labels on the mounting plate as well as the sections [12. Technical data](#) and [14. Appendix](#).

 **Max. 3 bar**
Min. 0.5 l/min

The maximum flow through the flow cell of the DID is limited by the integrated flow restrictor.

Use a vented outlet or a pressure-loading valve to avoid siphon effect. Do not install a shut-off valve in the outlet line.



4.3.1 Installing the measuring water inlet and outlet

1. Remove the blind plugs from the inlet thread and the outlet thread of the flow cell.



Fig. 1 Blind plug

2. Connect the Rp 1/2" hose adaptor fittings to the inlet and the outlet.
 - The Rp 1/2" hose adaptor fittings are included in the standard delivery of the DID.
3. Assemble the inlet and the outlet hose as follows:
 - Push union nut and tensioning ring across the hose.
 - Push the cone part fully into the hose, see fig. 2.
 - Attach the cone part with hose to the adaptor fitting.
 - Tighten the union nut manually. Do not use tools.

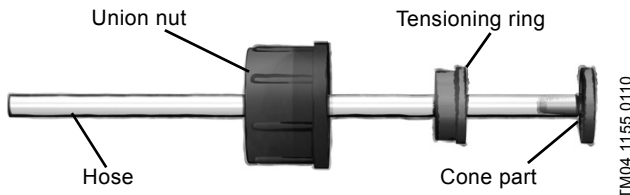


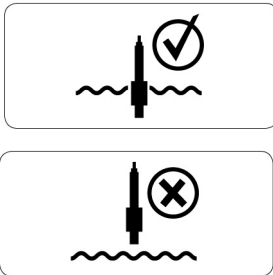
Fig. 2 Connecting the inlet hose

Avoid high loads caused by long or heavy hoses hanging from the outlet port of the flow cell. A strain-relief for the outlet hoses may be required.

4.4 Installing the sensors

All sensors used in the DID are packed separately to avoid damage. They have to be prepared before installation according to the "Installation" section in the manual "Sensors V1" supplied with the sensor.

- For installation of the sensors in the flow cell, see the manual "Sensors V1" supplied with the sensors.
- As soon as the sensors are ready for operation, they must be wetted.



- Sensors equipped with electrolyte and membrane cap must remain powered.

4.5 Checking the tightness

1. Carefully open the inlet valve and check for any leakage of the flow cell assembly.
2. Check if any air bubbles are visible within the hose.
3. Purge the bubbles out by temporary increase of the flow.

5. Starting up the product

5.1 Initial startup

WARNING

Electric shock



- Switch off the power supply before startup.
- IP65 only applies, if the housing cover and the terminal compartment of the CU 382 are properly closed, and the appropriate cable glands or dummy caps are fitted.
- Initial startup must be carried out by qualified and authorised persons.

1. Connect all sensors to the CU 382 control unit. See also section 6.2 *Connecting the sensors to the CU 382 control unit*.
2. Slowly open the inlet valve.
3. Check the hydraulic system of the DID for leakages.

6. Electrical connection

6.1 Terminal connection

WARNING

Electric shock



- Switch off the power supply of the DID and all devices you plan to connect before terminal connection.
- Terminal connection must be carried out by qualified and authorised persons.

6.1.1 Connecting the terminals

1. Remove the two cross-head screws on the terminal compartment of the CU 382 control unit.
2. Open the terminal compartment.
3. Connect the terminals according to the terminal connection plan shown in fig. 3.

N	90...240 VAC	2	4	P	90...240 VAC	1	3
B+	RS485	6	8	A-	RS485	5	7
+	Valve	10	12	+	12 V Out	9	11
+	Clean in	14	16	+	GND	13	15
'	4/20 #1	18	20	+	4/20 #1	17	19
'	4/20 #2	22	24	+	4/20 #2	21	23
'	4/20 #3	26	28	+	4/20 #3	25	27
'	4/20 In #1	30	32	PS	4/20 In #1	29	31
'	In #1	34	36	+	4/20 In #1	33	35
'	In #2	38	40	+	In #1	37	39
'	Flow #1	42	44	+	In #2	41	43
B+	RS485	46	48	+	Flow #1	45	47
MOD 1	MOD 1	50	52	A-	RS485	49	51
MOD 1	MOD 1	54	56	+	MOD 1	53	55
MOD 2	MOD 2	60	62	+	MOD 1	57	59
MOD 2	MOD 2	64	66	+	MOD 2	61	63
C	Relay 1	70	72	+	MOD 2	65	67
C	Relay 2	74	76	NON	Relay 1	69	71
C	Σ Err	80	82	NON	Relay 2	73	75
C	Σ Err	84	86	NON	Σ Err	77	79
				NON	Σ Err	81	83

Fig. 3 Terminal connection plan

Description and use of the terminals

Terminal	Description	Use	
1	Phase	Power supply (100-240 V, 50-60 Hz)	
2	Neutral		
3	PE		
4	PE		
5-8	Unused, can be used to connect S::CAN sensors		
9	12 V output	Cleaning valve control	
10	Cleaning valve output (provides power for the valve +12 V)		
11	GND, for cleaning valve and external trigger		
12	External trigger for cleaning valve		
13	4-20 mA output #1 (+)	Analog outputs	
14	4-20 mA output #1 (-)		
15	Shield connections (optional)		
16			
17	4-20 mA-output #2 (+)		
18	4-20 mA-output #2 (-)		
19	Shield connections (optional)		
20			
21	4-20 mA-output #3 (+)	Analog input	
22	4-20 mA-output #3 (-)		
23	Power supply for 4-20 mA input (only for passive sources)		
24	4-20 mA-input #1 (-)		
25	4-20 mA-input #1 (+)	Digital inputs	
26	Shield connection (optional)		
27	Digital input #1 (+), relay to switch on/off or as pulse source		
28	Digital input #1 (-)		
29	Digital input #2 (+), relay to switch on/off or as pulse source	Flow cell - flow switch	
30	Digital input #2 (-)		
31	Flow #1, flow switch of bypass flow cell, pre-wired in BF1/BF3	Modbus to upper controls (CU 382 is slave)	
32	Flow #1, flow switch of bypass flow cell, pre-wired in BF1/BF3		
33	RS485 A-		
34	RS485 B+		
35	RS485 GND	Alarm relay	
36	RS485 shield		
37-46	Unused, for later add-ons		
47	Relay 1-NO, switches on/off or provides pulse frequency		
48	Relay 1-C, common connector	Digital output (with relays)	
49	Relay 2-NO, switches on/off or provides pulse frequency		
50	Relay 2-C, common connector	Alarm relay	
51	Alarm relay NO Relay closes on alarm		
52	Alarm relay C, common connector		
53	Alarm relay NC Relay opens on alarm		
54	Alarm relay C, common connector, duplication of 52		

6.2 Connecting the sensors to the CU 382 control unit

All sensors for the DID have plugs.

Sensors for immersion have fixed cables with plugs.

1. Make sure that the sensor plug and the socket of the CU 382 control unit are dry and clean.
 - Otherwise communication errors may occur and the DID might be damaged.
2. Connect the sensor to a compatible socket on the CU 382 control unit using the connection cable provided.
3. Cover the connectors that are not used with protective caps.

6.3 Connecting the valve for automatic cleaning

Only DID with sensor for tank immersion with valve for automatic cleaning:

- Connect the valve to terminals 10 ("Valve", "12 VDC") and 11 (GND). See fig. 3.
- If an external cleaning signal is used, connect the signal line to terminal 12 ("Clean In", minimum 5 V).

6.4 Connecting further inputs and outputs

Observe the supplier manuals when connecting actuator devices, such as dosing pumps or regulators.

6.5 Connecting the CU 382 to the power supply

WARNING**Electric shock**

- Switch off the power supply before connecting the CU 382 control unit.
- IP65 only applies, if the housing cover and the terminal compartment of the CU 382 are properly closed, and the appropriate cable glands or dummy caps are fitted.
- Electrical connection must be carried out by qualified and authorised persons.

- Install an earth leakage circuit breaker for the power supply.
- If the installation is in an area prone to lightning strikes, install surge protection.
- Use an AC power supply with an earthed conductor wire.
 - PE: potential earth
- Connect the process medium (e.g. waste water) to the same earth ground with less than 0.5 Ohm.

6.6 Start-up of the operating software

If the CU 382 control unit is connected to the power supply, it starts up and shows the s::can logo for 5 seconds. At first start-up or if the internal settings have been reset to default, the CU 382 control unit starts with the selection of the language. See also section 9.3.7 *Select language....*

After this, the parameter screen is shown and the most recent information is displayed.

6.7 Sensor initialisation

Every sensor that is connected to the CU 382 control unit must be installed and initiated individually. This can be done using the software-supported initialisation process.

- Connect only the sensor you want to initialise to the CU 382.
 - An individual address is allocated to the initialised sensor.

If no sensor was connected before, the CU 382 jumps into the "Add MODBUS sensor..." menu to check for connected sensors, right after setting the language.

7. Storing and handling the product

- Before storing the DID, empty all pipes, hoses and flow cells, remove the sensors and pack them separately.
- Observe the permissible ambient conditions. See section 12. Technical data.
- Do not expose the DID to strong impact, mechanical loads or vibrations.
- Keep the DID away from corrosive substances, vapours of organic solvents, nuclear radiation and strong electromagnetic radiation.
- Use the original packaging or equivalent to protect the DID during storage.

8. Product introduction

8.1 Product description

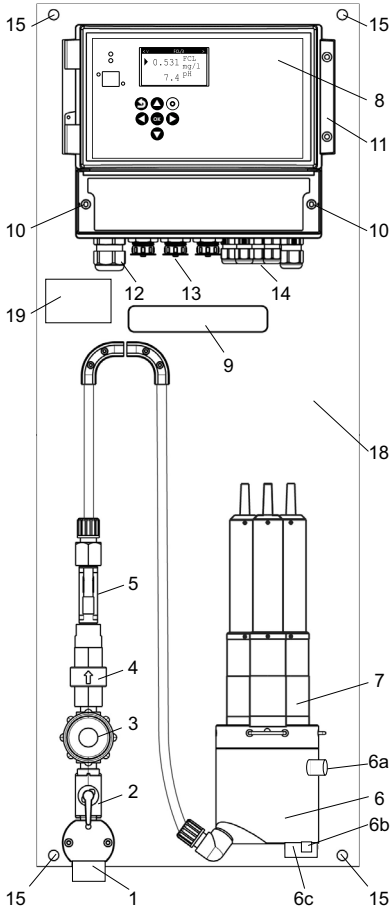


Fig. 4 Components of DID with flow cell

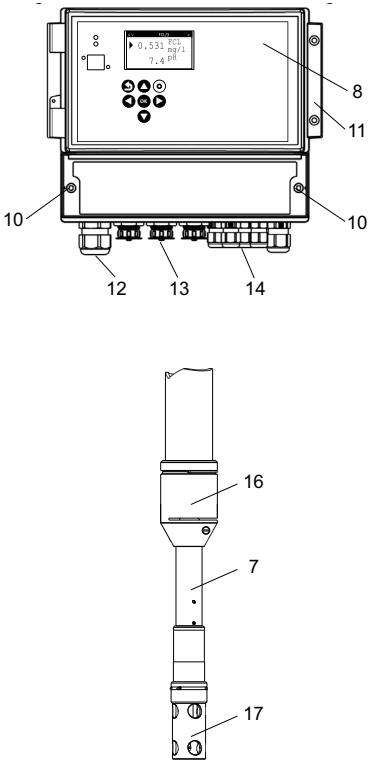


Fig. 5 Components of DID with immersed sensor

Pos.	Description
1	Water inlet
2	Inlet valve
3	Inlet strainer
4	Flow restrictor, mechanical
5	Flow detector, digital
6	Flow cell for up to three sensors
6a	Sampling valve
6b	Sampling point
6c	Water outlet
7	Sensors
8	CU 382 control unit
9	Opening for power supply and sensor cables
10	Screws for the terminal compartment
11	Latch of CU 382 cover
12	Cable gland for power supply
13	Connectors for Modbus sensors
14	Cable gland for signal inputs and outputs
15	Holes for wall mounting
16	Sensor holder
17	Protective cover
18	Mounting plate
19	DID nameplate

8.2 Intended use

The DID is designed for online monitoring of water quality parameters and offers control functions for these parameters.

The DID must be used in accordance with the instructions given in this manual. The values mentioned in this manual must be observed.

8.3 Functional principle

The CU 382 control unit has a display and 7 operating buttons on the front. The operating software starts automatically when the CU 382 is powered-on.

The CU 382 control unit collects and displays readings from all sensors via a digital bus connection. The CU 382 control unit includes a PID (Proportional-Integral-Derivative) controller. With the PID controller and dosing units, a closed control loop can be set up.

The hydraulic part of the DID manages the complete water flow from the inlet to the outlet. This includes filtration, pressure reduction, flow restriction and flow control. The DID comprises specifically designed flow cells for mounting of the sensors and appropriate sampling.

8.4 Identification

8.4.1 Nameplate

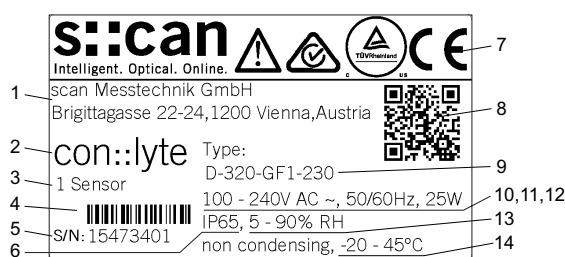


Fig. 6 Nameplate on the CU 382 control unit

Pos.	Description
1	Manufacturer and country of origin
2	s::can product name
3	Number of sensors
4	Bar code
5	Serial number (S/N)
6	Enclosure class
7	Certification marks
8	QR code
9	Type
10	Voltage [V]
11	Frequency [Hz]
12	Power consumption [W]
13	Humidity limits
14	Temperature limits

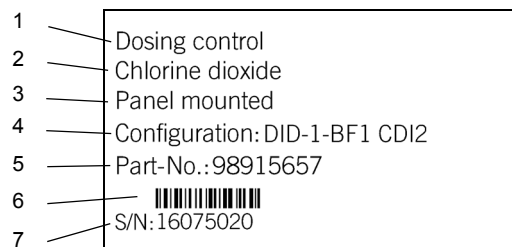


Fig. 7 Nameplate on the DID

Pos.	Description
1	s::can product name
2	Parameter
3	Mounting type
4	Type
5	Part number
6	Bar code
7	Serial number (S/N)

8.4.2 Type key

Example: DID-3 BF3-FCL2/TCL2/pH

	Code	Description
CU 382 control unit variants	DID-1	CU 382-1 for 1 parameter + temperature, 100-240 VAC, 50-60 Hz
	DID-3	CU 382-3 for 3 parameters + temperature, 100-240 VAC, 50-60 Hz
Installation variant	BF1	Bypass flow cell, 1 sensor
	BF3	Bypass flow cell, 3 sensors
	TI1	Tank immersion, 1 sensor
	TI2	Tank immersion, 2 sensors
Parameters/ Sensors	FCL2	Free chlorine 0-2 ppm
	FCL20	Free chlorine 0-20 ppm
	TCL2	Total chlorine 0-2 ppm
	TCL20	Total chlorine 0-20 ppm
	CDI2	Chlorine dioxide 0-2 ppm
	CDI20	Chlorine dioxide 0-20 ppm
	HP2	Hydrogen peroxide 0-200 ppm
	HP20	Hydrogen peroxide 0-2000 ppm
	PA2	Peracetic acid 0-200 ppm
	PA20	Peracetic acid 0-2000 ppm
	pH	pH 2-12 pH up to 10 bar and 70 °C
	ORP	ORP, -2000 mV to +2000 mV up to 10 bar, up to 70 °C
	CND	Conductivity, 1-500,000 µS/cm (2-42 PSU), up to 20 bar, up to 70 °C

Remarks:

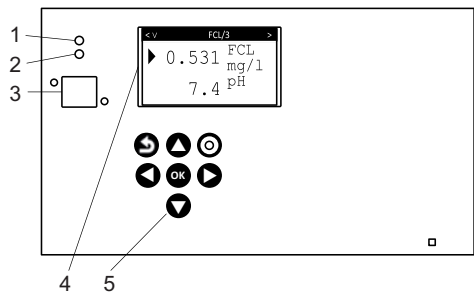
- All sensors include temperature measurement
- Disinfectant parameters (FCL, TCL, CDI, HP or PA) are only possible with BF1 or BF3
- BF1 and BF3 include flow detection
- BF1 and BF3 systems include 1.0 m sensor cables
- Installation variants TI1 and TI2 are only possible for parameters PH, ORP or CND
- TI1 and TI2 systems include 7.5 m sensor cable(s)
- Mains cable is generally not included, please select and order from accessories list.

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9. Operation

9.1 Operating elements



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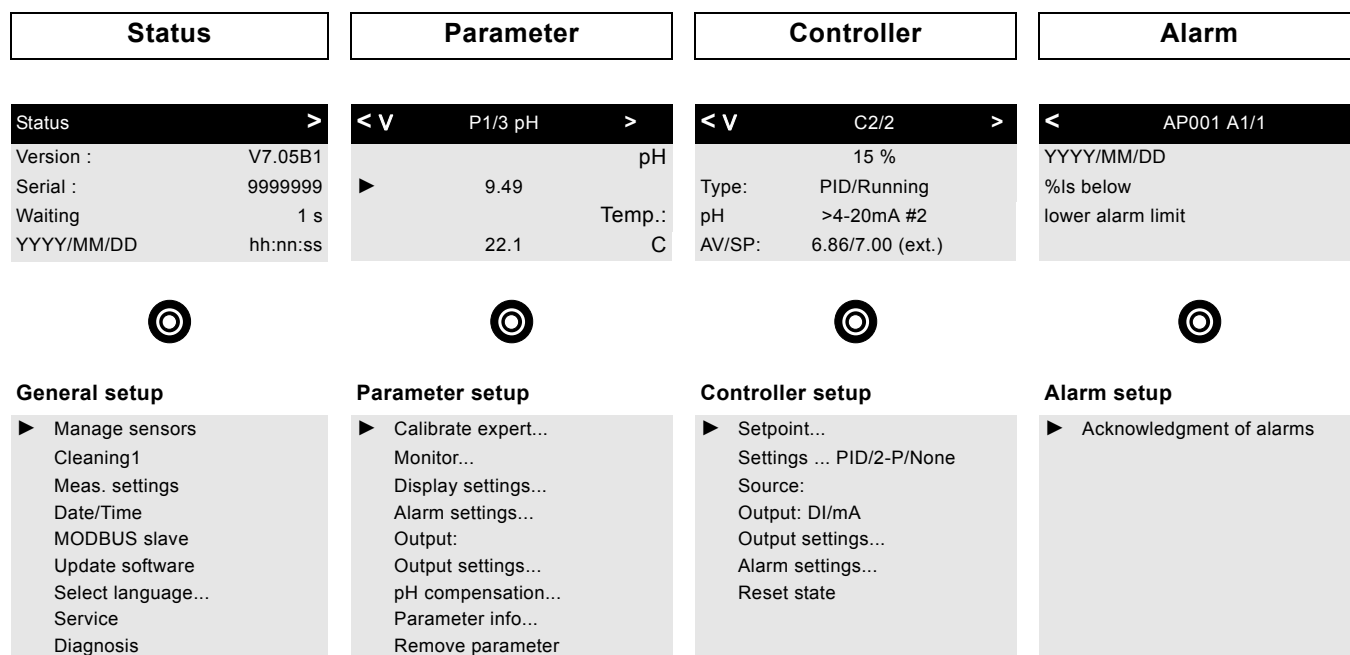
Fig. 8 CU 382 operating elements

Pos.	Description
1	Power LED, green
2	Communication LED, orange Flashes during communication with a sensor.
3	USB slot
4	Display
Operating buttons	
	[Back] button Exits the current menu
	[Up] button - Moves the marker to the line above - The first character of the selected line is a triangle - Increases values
	[Function] button - Enters the setup menus - Disables or deletes values
5	[Left] button Moves the marker to the left
	[OK] button - Enters the selected menu - Confirms the selected line or value - Acknowledges alarms
	[Right] button Moves the marker to the right
	[Down] button - Moves the marker to the line below - The first character of the selected line is a triangle - Decreases values

9.2 Software overview

The CU 382 has four different screens and menus, which can be selected by the [Right] and [Left] buttons. The default screen is the "Parameter" screen.

When pressing the [Back] button several times in a submenu, you come back to one of the following screens:



9.2.1 Status screen

Status	>
Version :	V7.05B1
Serial:	9999999
Waiting	1 s
2015/Oct/05	16:40:49

"Version :"	Software version
"Serial:"	Serial number of the CU 382
"Waiting"	Waiting time until the next action (start of measurement or start of cleaning) starts. The waiting time can be set in the "Meas. settings" menu.
"YYYY/MM/DD"	Current date and time

9.2.2 Parameter screen

Up to 8 parameters can be configured and displayed in the parameter screen, for example:

- Sensor reading
- Analog input
- Digital input
- Flow guard

Scroll through the list of configured parameters with the [Up] and [Down] buttons.

< V	P1/3 FCL	>
▶	0.531	FCL
		mg/l
	7.8	pH

"FCL"	Parameter name
"mg/l"	Parameter unit
"pH"	Parameter name

Calibration

"Span" calibration and "Offset" calibration can be started directly from the parameter screen. For the calibration procedures, see [9.4.1 Calibrate expert...](#)

9.2.3 Controller screen

Up to 3 controllers can be configured and displayed.
Scroll through the list of controllers with the [Up] and [Down] buttons.

< ^ C2/2 pH >	
15 %	
Type:	PID/Running
pH	>4-20mA #2
AV/SP:	6.86/7.00 (ext.)

"C2"	Controller 2
"15 %"	Output of the controller in %
"Type:"	Controller type: "2-P" or "PID" Controller status "Running": the controller is running and dosing is activated "Stopped": the controller is stopped and no dosing takes place For more information see section 9.5 Controller setup .
"pH"	Input parameter Input options: sensor reading, analog input, digital input
">4-20mA #2"	Output of the controller Output options: analog outputs (4-20mA), digital outputs (PULSE or PWM)
"AV/SP"	"AV" is the actual value (input parameter reading) "SP" is the setpoint (desired value of the input parameter reading) Source of the setpoint: "External" "Manual"

Adjusting the output manually

1. Press [OK].
2. Select a value with the [Up] and [Down] buttons.
3. Confirm with [OK].
 - The controller type changes to "Hand".
4. Press [OK] again to go back.
5. Press the [Function] button to clear the manual value of the output.
 - The value changes to "---".
6. Press [OK] to set the controller back to "2-P" or "PID".

9.2.4 Alarm screen

< AP001 A1/1 >	
2015/Oct/05	16:42:21
%ls below lower alarm limit	

"AP001 A1/1"	The amount of alarms and the number of the selected alarm
"2015/Oct/05"	The date of occurrence of the alarm
"16:42:21"	The time of occurrence of the alarm
"%ls below"	The alarm message
"lower alarm limit"	

An alarm can be acknowledged with [OK]. If the reason of the alarm is still present, a new alarm will be generated immediately.

9.3 General setup

In the "Setup" menu, general settings can be made.

- Press the [Function] button in the status screen to open the "Setup" menu.

v Setup	
▶ Manage sensors... - Cleaning 1... - Meas. settings... - Date/Time... - MODBUS slave...	

^ Setup	
▶ MODBUS slave... - Update software... - Select language... - Service... - Diagnosis...	

Submenu	Settings
"Manage sensors..."	Add, remove, configure sensors or parameters.
"Cleaning 1..."	DID with immersed sensor and cleaning device: Configure cleaning device.
"Meas. settings..."	Set the measurement interval between 1-99 s.
"Date/Time..."	Adjust date and time.
"MODBUS slave..."	If CU 382 is used as MODBUS-Slave: Change MODBUS settings.
"Update software..."	Update the software of the control unit via the USB port.
"Select language..."	Choose the operating language.
"Service..."	Make additional settings and function checks. This menu is protected by a password and should only be opened by an authorised person.
"Diagnosis"	Only for service, if the DID is sent back to the manufacturer.

9.3.1 Manage sensors

Manage sensors	
▶ pH::lyser - chlori::lyser - digital in #1 - Add sensor...	

"pH::lyser/0/1"	Names of the connected sensors
"chlori::lyser/0/2"	
"digital in #1"	Configured inputs (digital or analog)
"Add sensor..."	Add a new sensor

Sensor configuration

- Select a sensor or an input with the [UP] and [DOWN] buttons. Confirm with [OK].
 - The "Sensor configuration" menu opens.

Sensor configuration	
►	Configure...
	Add parameters...
	Remove

Configure...

- Select "Configure..." with the [Up] and [Down] buttons. Confirm with [OK].
 - The "Configure..." menu opens.

Configure...	
►	Type: 4-20mA

The "Configure..." menu is available for analog and digital inputs. If an analog input is chosen, the following settings can be configured:

- "0-20mA"
- "4-20mA"

Configure...	
►	Type: State
	Invert pol. : No

Configure...	
►	Type: Count
	Interval [s] : 60

If a digital input is chosen, the following settings can be configured:

- "State": Depending on the contact type of the digital input, the result is 1 or 0.
 - The contact type of the digital input can be changed with "Invert pol. :".
- "Count": Counts the digital pulses within the interval.
 - The interval can be defined with "Interval [s] :".

Adding a parameter

Add para.	
►	Add pH
	Add pH-mV
	Add Temp.:

- Select a parameter with the [Up] and [Down] buttons.
- Confirm with [OK].

Removing a sensor

- Select "Remove" with the [Up] and [Down] buttons.
- Confirm with [OK].

The sensor and the parameters for the sensor are removed. If an input of a controller is linked to the sensor or parameter, the controller is deactivated.

Adding a new sensor

- Select "Add sensor..." in the "Manage sensors" menu with the [Up] and [Down] buttons.
- Confirm with [OK].
 - The "Add new sensor" menu opens.

►	Add 0/4-20mA...
	Add digital in...
	Add MODBUS sensor...

- To add an analog sensor select "Add 0/4-20mA..." with the [Up] and [Down] buttons. Confirm with [OK].
- To add a digital sensor select "Add digital in..." with the [Up] and [Down] buttons. Confirm with [OK].
- To add a MODBUS sensor select "Add MODBUS sensor..." with the [Up] and [Down] buttons. Confirm with [OK].
 - The MODBUS port for a new sensor is searched and the new sensor is added to the sensor list automatically. No further interaction is required.

After adding a sensor, a parameter has to be added. See section [Adding a parameter](#).

9.3.2 Cleaning device

The "Cleaning 1..." menu is used for the DID with sensor for tank immersion, if an automatic cleaning device is installed.

Cleaning 1...	
►	Interval [s] : 120
	Duration[s] : 2
	Wait [s] : 0

"Interval [s] :" The time between two cleaning actions

"Duration[s] :" The time the cleaning device is activated for one cleaning action

"Wait [s] :" The time before starting a new measurement after a cleaning action is finished

Changing the cleaning interval

- Select "Interval [s] :" with the [Up] and [Down] buttons. Confirm with [OK].
- Change the interval value with the [Up] and [Down] buttons. Confirm with [OK].

Changing the cleaning duration

- Select "Duration[s] :" with the [Up] and [Down] buttons. Confirm with [OK].
- Change the duration value with the [Up] and [Down] buttons. Confirm with [OK].

Changing the delay before a new measurement

- Select "Wait [s] :" with the [Up] and [Down] buttons. Confirm with [OK].
- Change the delay value with the [Up] and [Down] buttons. Confirm with [OK].

9.3.3 Measurement settings

Meas. settings	
► Interval [s] :	5

"Interval [s] :" Time between two measurements in a range of 1-99 seconds.

Changing the measurement interval

1. Select "Interval [s] :" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the interval value with the [Up] and [Down] buttons. Confirm with [OK].
 - This changes also the sampling interval of the analog and digital inputs.

9.3.4 Date/Time

Date/Time	
► Year :	2016
Month :	Jan
Day :	8
Hour :	6
Minute :	43

Changing the year

1. Select "Year :" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the year value with the [Up] and [Down] buttons. Confirm the new year with [OK].

Changing the month

1. Select "Month :" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the month value with the [Up] and [Down] buttons. Confirm the new month with [OK].

Changing the day

1. Select "Day :" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the day value with the [Up] and [Down] buttons. Confirm the new day with [OK].

Changing the hour

1. Select "Hour :" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the hour value with the [Up] and [Down] buttons. Confirm the new hour with [OK].

Changing the minute

1. Select "Minute :" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the minute value with the [Up] and [Down] buttons. Confirm the new minute with [OK].

9.3.5 MODBUS slave

If the CU 382 is used as a Modbus slave, the connection settings can be configured in the "MODBUS slave" menu.

MODBUS slave	
► Address:	1
Parity:	Odd
Baudrate:	38400

"Address:" The Modbus ID. The address of each device must be unique in the Modbus environment.

"Parity:" The parity of the communication with RS 485 bus

"Baudrate:" The baud rate of the communication with RS 485 bus

Changing the Modbus address

1. Select "Address:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the address value with the [Up] and [Down] buttons. Confirm with [OK].

Changing the parity

1. Select "Parity:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parity with the [Up] and [Down] buttons. Confirm with [OK].

Changing the baud rate

1. Select "Baudrate:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the baud rate with the [Up] and [Down] buttons. Confirm with [OK].

9.3.6 Update software

The operating software of the CU 382 control unit can be updated in the field with a USB stick. See also section [9.3.6 Update software](#).

- To activate the software, press the [Function] button and keep it pressed.

Update software
Press FUNC key and keep pressed!

9.3.7 Select language...

Select language...
► English
Deutsch
Francais

Changing the operating language of the CU 382 control unit

1. Select the desired language with the [Up] and [Down] buttons.
2. Confirm the new language with [OK].

9.3.8 Service

Setup	
► Access code:	0001

Use the "Access code:" "0001" to open the "Service" menu.

v Service	
► 4-20mA Outputs	
0/4-20mA Inputs	
Internal SD-card	
Digital Outputs	
Digital Input	

^ Service v	
► Digital Input	
Internal Sensors	
USB	
Date/Time...	
Hardware	

^ Service	
► Date/Time...	
Hardware	
Serial number	
Communication test	
Modules	

The "Service" menu allows the following:

"4-20mA Outputs"	Test and recalibrate the 4-20 mA outputs
"0/4-20mA Inputs"	Test and recalibrate the 0/4-20 mA inputs
"Internal SD-card"	Format the SD card
"Digital Outputs"	Switch all digital outputs manually
"Digital Input"	Read in all digital inputs
"Internal Sensors"	Read in all internal sensors
"USB"	Measure the performance of the USB stick
"Date/Time..."	Set the time and the date
"Hardware"	Read out the hardware revision
"Serial number"	Serial number of the CU 382
"Communication test"	Check the communication with the sensors and external devices
"Modules"	Check the integrated plug-in modules

9.4 Parameter setup

In the parameter setup menu, a parameter can be calibrated, monitored, linked to an alarm, linked to an output or removed.

v P1 FCL/mg/l	
► Calibrate expert...	
Monitor...	
Display settings...	
Alarm settings...	
Output:	4-20mA #1

^ P1 FCL/mg/l	
► Output:	4-20mA #1
Output settings...	
pH compensation...	
Parameter info...	
Remove parameter	

Submenu	Settings
"Calibrate expert..."	Calibrate a parameter
"Monitor..."	Display the raw value and the status of the parameter and the sensor
"Display settings..."	Configure the appearance of the parameter
"Alarm settings..."	Configure the alarm-trigger of the parameter
"Output:"	Allocate an output to a parameter value
"Output settings..."	Configure the selected output for the parameter
"pH compensation..."	Enable or disable the software pH compensation for the chlorine sensor
"Parameter info..."	Display a summary of the parameter settings
"Remove parameter"	Remove the parameter

9.4.1 Calibrate expert...



Before calibration make sure that the parameter value is stable.

Depending on the connected sensor, several types of calibration can be performed.

- "Local" is set by default. "Local" calibration is performed by the user during operation.
- "Global" is used for factory calibration. "Global" is performed by the manufacturer of the sensors.

Depending on the sensor type, different "Local" calibration modes can be chosen in "Mode:":

- "Offset"
- "Linear"
- "Span"

Sensor type	Parameter	Mode:
pH::lyser, redo::lyser	pH, ORP	Offset, Linear
chlori::lyser chlodi::lyser, peroxy::lyser, hyper::lyser	Cl ₂ , ClO ₂ , H ₂ O ₂ , PAA	Span
condu::lyser	Conductivity	Span

See the manual "Sensors V1" for more detailed information.

"Offset" calibration

v P2 Calibrate expert...	
► Type:	Local
Mode:	Offset
Perform Calibration	
Value:	7.77
Lab 1:	7.8

^ P2 Calibrate expert...	
► Perform Calibration	
Value:	7.77
Lab 1:	7.8
Sample 1 :	54.34
Offset	-1.85

"Type:"	Calibration type: "Local" (default) or "Global" (for factory calibration)
"Mode:"	Calibration mode: "Offset", "Linear", "Span"
"Value:"	The value measured in the current sensor calibration
"Lab 1:"	The value received from the analysis of the sample with a reference method
"Sample 1 :"	The raw value of the sample measured by the sensor. The raw value can have a different unit.

Performing "Offset" calibration

The "Offset" calibration is an in-situ calibration. The sensor must not be removed from the flow cell.

1. Select "Offset" in the "Mode:" entry field with the [Up] and [Down] buttons. Confirm with [OK].



Before taking a sample for calibration, flush the sampling point 3-4 times with approximately 10 ml each time.

2. Take a sample of approx. 10 ml.
3. While taking the sample, select "Sample 1 :" with the [Up] and [Down] buttons. Confirm with [OK].
 - The raw value of the sample is measured and stored in the sensor.
4. Analyse the sample with a reference method and enter the result into the "Lab 1:" field with the [Up] and [Down] buttons.
5. Select "Perform Calibration" with the [Up] and [Down]. Confirm with [OK].
 - The calibration process starts.

The calibration result is shown in "Value:".

The calibration coefficient is shown in "Offset", and stored in the sensor.

Performing "Linear" calibration

1. Select "Linear" in the "Mode:" entry field. Confirm with [OK].
2. Take the sensor out of the flow cell or the tank.
3. Immerse the sensor into a beaker with the first reference solution for "Sample 1 :".
4. Let the sensor adapt to the solution until the displayed value is stable, before measuring the raw value.
5. Read the value on the label of the first reference solution. Select the value in the "Lab 1:" field with the [Up] and [Down] buttons. Confirm with [OK].
 - Use a temperature-compensated value.
 - The entire measuring range that can be expected at the location should be covered between "Sample 1 :" and "Sample 2 :".

6. Select "Sample 1 :" with the [Up] and [Down] buttons. Confirm with [OK].
 - The measured raw value is shown and stored in the sensor.
7. Carefully rinse the sensor with distilled water three times before measuring the second value.
8. Immerse the sensor into a beaker with the second reference solution for "Sample 2 :".
9. Let the sensor adapt to the solution until the displayed value is stable, before measuring the raw value.
10. Read the value on the label of the second reference solution. Select the value in the "Lab 2:" field with the [Up] and [Down] buttons. Confirm with [OK].
 - Use a temperature-compensated value.
11. Select "Sample 2 :" with the [Up] and [Down] buttons. Confirm with [OK].
 - The measured raw value is shown and stored in the sensor.
12. Select "Perform Calibration" with the [Up] and [Down] buttons. Confirm with [OK].
 - The calibration process starts.

The calibration result is shown in "Value:".

The calibration coefficients are shown in "Offset" and "Slope", and stored in the sensor.

"Span" calibration

v P1 Calibrate expert...	
► Type:	Local
Mode:	Span
Perform Calibration	
Value:	0.50
Private:	3.0

^ P1 Calibrate expert...	
► Value:	0.50
Private:	3.00
Lab 1:	0.50
Sample 1 :	3.00
Slope	6.00

"Type:"	Calibration type: "Local" (default) or "Global" (for factory calibration)
"Mode:"	Calibration mode: "Offset", "Linear", "Span"
"Value:"	The value measured in the current sensor calibration
"Private:"	Internal sensor value which is only relevant for service. If a pH::lyser or redo::lyser is connected, this value is the quality factor of the sensor. 0 means bad quality and 1 means perfect quality. If another sensor connected, this value is the currently measured raw value of the sensor.
"Lab 1:"	The value received from the analysis of the sample with a reference method can be entered.
"Sample 1 :"	The raw value of the sample measured by the sensor is displayed. The raw value can have a different unit.
"Slope"	The calibration coefficient

Performing "Span" calibration

The "Span" calibration is an in-situ calibration. The sensor must not be removed from the flow cell.

1. Select "Span" in the "Mode:" entry field with the [Up] and [Down] buttons. Confirm with [OK].



Before taking a sample for calibration, flush the sampling point 3-4 times with approximately 10 ml each time.

2. Take a sample of approx. 10 ml.
3. While taking the sample, select "Sample 1 : " with the [Up] and [Down] buttons. Confirm with [OK].
 - The raw value of the sample is measured and stored in the sensor.
4. Analyse the sample with a reference method and enter the result into the "Lab 1: " field with the [Up] and [Down] buttons.
5. Select "Perform Calibration" with the [Up] and [Down] buttons. Confirm with [OK].
 - The calibration process starts.

The calibration result is shown in "Value:".

The calibration coefficient is shown in "Slope", and stored in the sensor.

9.4.2 Monitor...

The "Monitor..." menu shows the raw value and status information of the chosen parameter and sensor. The operator can identify problems with the sensors in the "Monitor..." menu.

v P2 Monitor...	
► Value:	7.8
Private:	0.99
S-Status:	0000
S-Status(Pri):	0000
P-Status:	0000

^ P2 Monitor...	
► Private:	0.99
S-Status:	0000
S-Status(Pri):	0000
P-Status:	0000
P-Status(Pri):	0000

"Value:"	The value measured in the current sensor calibration
"Private:"	Internal sensor value which is only relevant for service. If a pH::lyser or redo::lyser is connected, this value is the quality factor of the sensor. 0 means bad quality and 1 means perfect quality. If another sensor connected, this value is the currently measured raw value of the sensor.
"S-Status:"	Sensor status (see manual "Sensors V1"). If "0000" is displayed, the status is ok.
"P-Status:"	Parameter status (see manual "Sensors V1"). If "0000" is displayed, the status is ok.

9.4.3 Display settings...

In the "Display settings..." menu the appearance of the parameter on the parameter screen and in the stored parameter files can be configured.

P2/Display settings...	
► Name:	pH 1
Unit:	
Disp. Format:	2
Load Defaults	

"Name:": Name of the parameter

"Unit:": Unit of the parameter

"Disp. Format:": Number of decimal places that are displayed

"Load Defaults" Reload default display settings

Changing the name of the parameter

1. Select "Name:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the name of the parameter with the [Up], [Down], [Left] or [Right] buttons. Confirm with [OK].

Changing the unit of the parameter

1. Select "Unit:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the unit of the parameter with the [Up], [Down], [Left] or [Right] buttons. Confirm with [OK].

Changing the number of decimal places

1. Select "Disp. Format:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the number of decimal places with the [Up] and [Down] buttons. Confirm with [OK].

Reloading the default display settings

1. Select "Load Defaults" with the [Up] and [Down] buttons. Confirm with [OK].
2. The default display settings are loaded from the sensor.
 - All modifications of the display settings are documented in the logbook of the CU 382 control unit.

9.4.4 Alarm settings...

In the "Alarm settings..." menu a lower and upper limit for an alarm can be defined. If the upper or lower limit is exceeded, an alarm is generated. Depending on the controller settings, the alarm can stop the controller.

- If in the "Alarm settings..." menu for the controller "Stop on Alarm:" is set to "Yes", and "Require ack.:" is set to "No", the controller starts running again automatically, when the parameter is back in the limits.
- If in the "Alarm settings..." menu for the controller "Stop on Alarm:" is set to "Yes", and "Require ack.:" is set to "Yes", the alarm must be confirmed manually to start the controller again, when the parameter is back in the limits.

See also section [9.5.5 Alarm settings...](#)

P2 Alarm settings...	
► Require ack.:	Yes
Al. lower:	6.5
Al. upper:	7.5
Output:	None
Use Sys-Error:	No

"Require ack.:" Acknowledgement of an alarm

"Al. lower:" Lower limit for an alarm

"Al. upper:" Upper limit for an alarm

"Output:" Output of an alarm

"Use Sys-Error:" Output of an alarm on the error relay

Changing the requirement for acknowledging an alarm

1. Select "Require ack.:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "Yes" or "No" with the [Up] and [Down] buttons. Confirm with [OK].

Changing the lower limit for triggering an alarm

1. Select "Al. lower:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the lower limit value with the [Up], [Down], [Left] or [Right] buttons. Confirm with [OK].

Changing the upper limit for triggering an alarm

1. Select "Al. upper:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the upper limit value with the [Up], [Down], [Left] or [Right] buttons. Confirm with [OK].

Defining the output of an alarm

1. Select "Output:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "None", "DigOut #1" or "DigOut #2" with the [Up] and [Down] buttons. Confirm with [OK].
3. If "DigOut #1" or "DigOut #2" is selected, the submenu "Invert pol. :" is shown.
 - The contact type of the digital output can be changed with "Invert pol. :".

Defining if the error relay is activated on an alarm

1. Select "Use Sys-Error:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "Yes" or "No" with the [Up] and [Down] buttons. Confirm with [OK].

9.4.5 Outputs

The following outputs can be selected:

- Analog output: "4-20mA #1", "4-20mA #2", "4-20mA #3"
- Digital output: "Level", "PULSE", "PWM"

Changing the output

1. Select "Output:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the output with the [Up] and [Down] buttons. Confirm with [OK].

9.4.6 Output settings...

In the "Output settings..." menu a parameter can be allocated to an analog output or a digital output.

Analog outputs

P2 Output settings...	
► [4mA]:	0.00
[20mA]:	14.00

"[4mA]:" The parameter value for 4 mA

"[20mA]:" The parameter value for 20 mA

Parameter values between these two limits are calculated according to a linear calibration.

Assigning the parameter value for 4 mA

1. Select "[4mA]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Assigning the parameter value for 20 mA

1. Select "[20mA]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Digital outputs

The following output types can be defined:

- Level
- PULSE
- PWM

Level

P2 Output settings...	
► Type:	Level
Threshold:	7.00

"Threshold:" "Threshold:" defines the limits for 0 or 1. Below the defined threshold the output is 0 (Relay NO = open). Above the defined threshold the output is 1 (Relay NO = closed).

Changing the digital output to "Level"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "Level" with the [Up] and [Down] buttons. Confirm with [OK].
 - An additional row is shown.

Changing the threshold value

1. Select "Threshold:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the threshold value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

PULSE

P2 Output settings...	
► Type:	PULSE
[0%]:	0.00
[100%]:	14.00
Pulses [Min]:	180

"[0%]:" Parameter value for 0 % output (0 pulses/min).

"[100%]:" Parameter value for 100 % output (maximum pulses/min).

"Pulses [Min]:" Maximum number of pulses per minute for 100 % output. The highest number is 180.

Changing the digital output to "PULSE"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "PULSE" with the [Up] and [Down] buttons. Confirm with [OK].
– Three additional rows are shown.

Assigning the parameter value for 0 % output

1. Select "[0%]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter value for 0 % output with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Assigning the parameter value for 100 % output

1. Select "[100%]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter value for 100 % output with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Changing the maximum of pulses/min for 100 % output

1. Select "Pulses [Min]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the maximum pulses/min value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

PWM

P2 Output settings...	
► Type:	PWM
[0%]:	0.00
[100%]:	14.00
Period [s]:	100

"[0%]:" Parameter value for 0 % output (0 pulses/min).

"[100%]:" Parameter value for 100 % output (maximum pulses/min).

"Period [s]:" Duration of one pulse for 100 % output. The highest duration is 100 s.

Changing the digital output to "PWM"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "PWM" with the [Up] and [Down] buttons. Confirm with [OK].
– Three additional rows are shown.

Assigning the parameter value for 0 % output

1. Select "[0%]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter value for 0 % output with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Assigning the parameter value for 100 % output

1. Select "[100%]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter value for 100 % output with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Changing the duration of one pulse for 100 % output

1. Select "Period [s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the value for the duration of one pulse with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

9.4.7 Parameter info...

The "Parameter info..." screen shows a summary of all important settings and information of one parameter.

v P2 Parameter info...	
► Sen.:	pH::lyser/0/
Name:	pH
Unit:	
Disp. Format:	2
P. lower:	0.00

^ P2 Parameter info...	
► Disp. Format:	2
P. lower:	0.00
P. upper:	14.00
Al. lower:	6.5
Al. upper:	7.5

"Sen.:" Name of the sensor

"Name:" Name of the parameter

"Unit:" Unit of parameter

"Disp. Format:" Number of decimal places that are displayed

"P. lower:" Lower end of the parameter measuring range

"P. upper:" Upper end of the parameter measuring range

"Al. lower:" Lower limit for triggering an alarm

"Al. upper:" Upper limit for triggering an alarm

9.4.8 pH compensation...

If the pH of the water is above 8.5, it is recommended to use the automatic pH compensation for the chlorine sensor. The pH compensation is only available for the chlorine sensor and requires a pH sensor.

P1 pH compensation...	
► Type:	Disabled

P1 pH compensation...	
► Type:	Auto
Source:	pH

"Type:" Compensation type

"Source:" The source for pH compensation (parameter pH from pH sensor)

"Auto" Automatic pH compensation

Enabling the pH compensation

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the type to "Auto" with the [Up] and [Down] buttons. Confirm with [OK].
 - The pH compensation is enabled.

Defining the source for pH compensation

1. Select "Source:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the source to the desired parameter (pH) with the [Up] and [Down] buttons. Confirm with [OK].
 - pH is defined as the source parameter.

9.4.9 Removing a parameter

1. Select "Remove" with the [Up] and [Down] buttons.
2. Confirm with [OK].
3. Enter the access code 0001 with the [Up] and [Down] buttons.
4. Confirm with [OK].
 - The parameter for the sensor is removed.
 - If an input of a controller is linked to the parameter, the controller is deactivated.

9.5 Controller setup

In the controller setup menu, a setpoint can be defined, controller parameters can be configured, the input for the controller can be chosen, the output for the controller value can be chosen, the digital output can be configured, alarms can be configured and the state of the controller can be reset.

v	C2 pH/
►	Setpoint:
	Settings ... PID
	Source: pH
	Output: DigOut #1
	Output settings...

^	C2 pH/	v
►	Source:	pH
	Output: DI/mA	DigOut #1
	Output settings...	
	Alarm settings...	
	Reset state	

Submenu	Settings
"Setpoint:"	The setpoint of the controller can be set.
"Settings ... PID/2-P"	The parameter for a PID or a 2-P controller can be configured.
"Source:."	The input parameter of the controller can be selected.
"Output settings..."	The output of the controller can be configured. Only visible if a digital output is defined.
"Alarm settings..."	The alarms for the controller can be configured.
"Reset state"	The state of the controller can be set to 0.

9.5.1 Setpoint...

The setpoint of the controller can be defined directly in the operating software of the controller, or an external setpoint adjustment can be configured. A setpoint can be controlled by an analog input or a digital input.

Two types of setpoint adjustment can be defined:

- "Manual"
- "External"

v	C2
►	Type: Manual
	Setpoint: 7.00

Changing the type to "Manual"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "Manual" with the [Up] and [Down] buttons. Confirm with [OK].

Changing the setpoint of the controller

1. Select "Setpoint:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the setpoint with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

The unit of the setpoint is defined by the parameter that is used as an input.

v	C2
►	Type: External
	Source: 4-20mA #1

Changing the type to "External"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "External" with the [Up] and [Down] buttons. Confirm with [OK].

Changing the source of the external setpoint

1. Select "Source:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the source with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

9.5.2 Settings ... PID

v	C2 settings
►	Type: PID
	Kp [%]: 5.0
	Ti [s]: 200.00
	Td [s]: 0.00
	Disturb.: DI 1

^	C2 settings
►	Td [s]: 0.00
	Disturb.: DI 1
	D. factor: 50.0
	Direction: up
	Deadband: 0.05

- "Kp [%]:" The proportional part of the PID controller with a range of 0 to 100 %
- "Ti [s]:" The integral part of the PID controller with a range of 1 to 3000 seconds.
- "Td [s]:" The derivative part of the PID controller with a range of 0 to 1000 seconds.
- "Disturb.:" Defines the source of the disturbance control.
- "D. factor:" The disturbance factor is a multiplicative factor
- "Direction:" Direction of the controller:
- "up": The controller increases the controller output, if the setpoint is above the parameter, and decreases the controller output, if the setpoint is below the parameter.
 - "down": The controller decreases the controller output, if the setpoint is above the parameter, and increases the controller output, if the setpoint is below the parameter.
- "Deadband:" A dead band is defined to avoid repeated activation-deactivation cycles (Hunting).

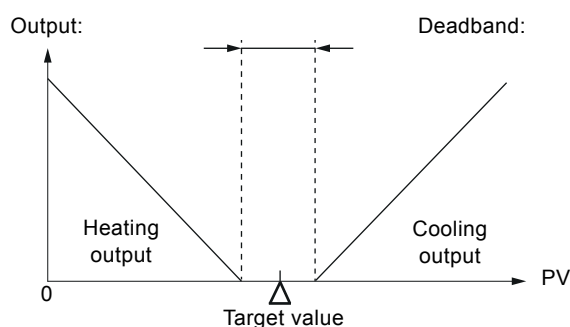


Fig. 9 Hunting (repeated activation-deactivation cycles)

Changing the controller type

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "PID", "2-P" or "None" with the [Up] and [Down] buttons. Confirm with [OK].
 - "None" deactivates the controller.

Changing the proportional part of the PID controller

1. Select "Kp [%]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the proportional part value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Changing the integral part of the PID controller

1. Select "Ti [s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the integral part value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Disabling the integral part of the PID controller

1. Select "Ti [s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Press the [Function] button and confirm with [OK] again to disable "Ti [s]:".
 - The displayed value is "+++++.++".

Changing the derivative part of the PID controller

1. Select "Td [s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the derivative part with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Disabling the derivative part of the PID controller

1. Select "Td [s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Press the [Function] button and confirm with [OK] again to disable "Td [s]:".
 - The displayed value is "0.00".

Disturbance control

The function of disturbance control (compound-loop control) is defined by following equation:

Controller output value with disturbance control = controller output value * Source value * "D. factor:".

Example:

- The controller output value is 50 %
- The source is the analog input with 1.2 mA
- The disturbance factor is 0.5

The new controller output is:

Controller output value with disturbance control = 50 [%] * 1.2 [mA] * 0.5 [1/mA] = 30 [%]

Assigning the disturbance control to an input (parameter)

1. Select "Disturb.:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the input for disturbance control with the [Up] and [Down] buttons. Confirm with [OK].

Changing the disturbance factor

1. Select "D. factor:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the disturbance factor with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Changing the direction of the controller

1. Select "Direction:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "up" or "down" with the [Up] and [Down] buttons. Confirm with [OK].

Changing the level of the deadband

1. Select "Deadband:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the value for the level of the deadband with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

9.5.3 Settings ... 2-P

v	C2 settings
►	Type: 2-P
	Hysteresis: 0.05
	Disturb.: DI 1
	D. factor: 50.0
	Direction: up

- "Type:" Controller type: "PID" or "2-P"
- "Hysteresis:" Hysteresis for the 2-P-controller. The switching points for the controller are:
Setpoint ± Hysteresis
- "Disturb.:" Defines the source of the disturbance control.
- "D. factor:" The disturbance factor is a multiplicative factor
- "Direction:" Direction of the controller:
- "up": The controller increases the controller output, if the setpoint is above the parameter, and decreases the controller output, if the setpoint is below the parameter.
 - "down": The controller decreases the controller output, if the setpoint is above the parameter, and increases the controller output, if the setpoint is below the parameter.

Changing the controller type

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "PID", "2-P" or "None" with the [Up] and [Down] buttons. Confirm with [OK].
 - "None" deactivates the controller.

Changing the hysteresis of the controller

1. Select "Hysteresis:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the hysteresis value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Assigning the disturbance control to an input (parameter)

1. Select "Disturb.:" with the [Up] and [Down] buttons. Confirm with [OK].
 - "Disturb.:" defines the source of the disturbance control. Analog outputs ("4-20mA") or digital outputs ("PULSE" or "PWM") can be selected.
2. Change the input for disturbance control with the [Up] and [Down] buttons. Confirm with [OK].

Changing the disturbance factor

1. Select "D. factor:" with the [Up] and [Down] buttons. Confirm with [OK].
 - "D. factor:" is a multiplicative factor.
2. Change the disturbance factor with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

Changing the direction of the controller

1. Select "Direction:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "up" or "down" with the [Up] and [Down] buttons. Confirm with [OK].

9.5.4 Output settings...

If an analog output "4-20mA #1", "4-20mA #2" or "4-20mA #3" is selected, the 0-100 % output of the controller is automatically assigned to 4-20 mA.

If a digital output is selected, two different types can be defined: "PULSE" or "PWM".

C2 Output settings...	
► Type:	PULSE
Pulses [Min]:	180

"Pulses [Min]:" Maximum number of pulses per minute for 100 % output. The highest number is 180.

Changing the digital output to "PULSE"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "PULSE" with the [Up] and [Down] buttons. Confirm with [OK].
 - An additional row is shown.

Assigning the parameter value for an output

The output range of the controller is 0 % to 100 %, so it is not necessary to assign the parameter value for an output.

The only parameter that can be changed is the maximum of pulses/min for 100 % output.

Changing the maximum of pulses/min for 100 % output

1. Select "Pulses [Min]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the maximum pulses/min value with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

C2 Output settings...	
► Type:	PWM
Period [s]:	100

"Period [s]:" Duration of one pulse for 100 % output. The highest duration is 100 s.

Changing the digital output to "PWM"

1. Select "Type:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "PWM" with the [Up] and [Down] buttons. Confirm with [OK].
 - One additional row is shown.

Assigning the parameter value for an output

The output range of the controller is 0 % to 100 %, so it is not necessary to assign the parameter value for an output.

The only parameter that can be changed is the duration of one pulse for 100 % output.

Changing the duration of one pulse for 100 % output

1. Select "Period [s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the value for the duration of one pulse with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

9.5.5 Alarm settings...

C2 Alarm settings...	
► Stop on Alarm:	
Stop on Error:	Yes
Max. 100%[s]:	200

"Stop on Alarm:" The behaviour of the controller can be defined for every parameter alarm.

"Stop on Error:" If "Stop on Error:" is "Yes", the controller stops, if a sensor error occurs. The status of the controller changes to "Stopped".

"Max. 100%[s]:" Maximum dosing time with a controller output of 100 %.

Changing the behaviour of the controller if an error occurs

1. Select "Stop on Error:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Select "Yes" or "No" with the [Up] and [Down] buttons. Confirm with [OK].

Changing the maximum dosing time

1. Select "Max. 100%[s]:" with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the maximum dosing time with the [Up], [Down], [Left] and [Right] buttons. Confirm with [OK].

9.5.6 Stop on Alarm:

v C2 Stop on Alarm:		
► FCL:	-----	
pH:		Up/Low
Temp.:		Upper

"FCL"	"-----"	The controller does not react or stop on an FCL alarm
"pH"	"Up/Low"	The controller stops, if the lower or upper alarm limit is reached.
"Temp.:"	"Upper"	The controller stops, if the upper alarm limit is reached.

The behaviour of the controller can be defined for every parameter alarm separately.

Changing the behaviour of the controller

1. Select the desired parameter with the [Up] and [Down] buttons. Confirm with [OK].
2. Change the parameter alarm value with the [Up] and [Down] buttons. Confirm with [OK].
 - The controller stops when the chosen alarm occurs.

9.5.7 Reset state

Resetting the state of the controller

1. Select "Reset state" with the [Up] and [Down] buttons.
2. Confirm with [OK] to reset the state of the controller.
 - The controller value is set to 0 %.
 - For the PID controller, all three parts are set to 0 (P = 0, I = 0, D = 0).

9.6 Alarm setup

< AP001 A1/1		
2015/Oct/05		16:42:21
%ls above		
lower alarm limit		

"AP001 A1/1"	The amount of alarms and the number of the selected alarm
"2015/Oct/05"	The date of occurrence of the alarm
"16:42:21"	The time of occurrence of the alarm
"%ls above"	
"lower alarm limit"	The alarm message

An alarm can be acknowledged with [OK]. If the reason of the alarm is still present, a new alarm will be generated immediately.

9.7 USB menu / Data transfer

The USB menu is activated when an USB stick is plugged in the USB slot.

USB data transfer	
►	Copy results
	Delete results
	Copy logfile
	Delete logs
	Update software

"Copy results"	All data regarding the parameters and controllers are copied to the USB stick.
"Delete results"	All data regarding the parameters and controllers are deleted.
"Copy logfile"	All data regarding the system logfiles are copied to the USB stick.
"Delete logs"	All system logfiles are deleted.
"Update software"	The software of the CU 382 control unit can be updated in the field with an USB stick. A file called "image.hex" has to be in the root directory of the USB stick. If the software update was successful, a message appears and the CU 382 control unit restarts.

Copying all data to the USB stick

1. Select "Copy results" with the [Up] and [Down] buttons.
2. Confirm with [OK] to copy all data to the USB stick.
3. The copied data are stored in a result file in .txt format. See [9.7.1 Result file](#).

Deleting all data

1. Select "Delete results" with the [Up] and [Down] buttons.
2. Confirm with [OK] button to delete the data.

Copying all logfiles to the USB stick

1. Select "Copy logfile" with the [Up] and [Down] buttons.
2. Confirm with [OK] to copy the data to the USB stick.

Deleting all logfiles

1. Select "Delete logs" with the [Up] and [Down] buttons.
2. Confirm with [OK] to delete the logfiles.

9.7.1 Result file

The result file is in .txt format and structured as follows:

Timestamp	Seconds since 1970/1/1
Time	YearMonthDay-HourMinuteSecond
P	Marks the beginning of the parameter data block
Parameter value	Value of the parameter
Parameter status	Status of the parameter
Parameter private status	Private status of the parameter
Upper alarm triggered	If upper alarm was triggered = "U" else "-"
Lower alarm triggered	If lower alarm was triggered = "L" else "-"
C	Marks the beginning of the controller data block
Controller output [%]	Output of the controller
Disturbance input	Input of disturbance control
Controller status	Status of the controller 0x0001 - General error 0x0002 - Input error 0x0004 - Output error 0x0008 - Disturbance error
Controller operating mode	Operating mode of the controller 0 - "Stopped" 1 - "Running" 2 - "Hold"

10. Maintenance

WARNING

Electric shock



- Switch off the power supply before connecting the power supply cable and relay contacts.
- Do not dismantle the control unit.
- Maintenance and repair must only be carried out by qualified persons.
- Observe the local safety regulations.

WARNING

Toxic material



- Parts of the DID can be contaminated with dangerous chemicals or pathogenic germs.
- If the DID has been in contact with dangerous chemicals or pathogenic germs, wear protective clothing and take all necessary precautions to prevent endangering your health when installing or dismantling the DID.

10.1 Cleaning

The following components of the DID require periodical checks and cleaning, the interval depends on the quality of the water in which they are used. We suggest to check these components on a monthly basis.

10.1.1 Inlet strainer

The strainer includes a filter that collects coarse material and can foul. Therefore, a periodical check of the strainer is necessary. We recommend to clean the filter in the following intervals:

- monthly
- if the flow drops, although all valves are open, the pump (if installed) is working properly and the inlet pressure is suitable.

Cleaning the inlet strainer

1. Turn off the water flow.
2. Unscrew the cap from the strainer.
3. Remove the filter.
4. Rinse the filter with drinking water.
5. Remove any coarse materials from the strainer housing.
6. Reinsert the filter.
7. Screw the cap back onto the strainer.

10.1.2 Flow detector

The flow detector contains inner parts that can be affected by fouling. In case the flow detector does not respond to flow changes, it can be dismantled from the DID and carefully cleaned with water and a soft brush.

Dismantling the flow detector

1. Turn off the water flow.
2. Disconnect the CU 382 control unit from the power supply.
3. Unscrew the hose connection.
4. Remove the hose.
5. Disconnect the wires of the flow detector from the CU 382.
6. Unscrew the flow detector.
7. Clean the flow detector carefully with water and a soft brush.
8. Screw in the flow detector back.
9. Reconnect the wires of the flow detector to the CU 382.
10. Refit the hose.
11. Screw on the hose connection.

10.1.3 Sensor

See the manual "Sensors V1" supplied with the sensor.

10.1.4 Control unit CU 382

If necessary, clean the surfaces of the CU 382 control unit with a damp and clean cloth.

10.2 Functional check

A functional check can be required for one of the following reasons:

- Initial startup
- Routine
- Malfunction
- Modification of the DID setup (e.g. integration of an additional sensor)
- Change of measuring location.

We recommend to perform a regular functional check as described in the table below. The functional check should be carried out monthly or weekly, depending on the application (water quality), the sensors connected and the environmental conditions.

For the functional check of the sensors, see the manual "Sensors V1" supplied with the sensor.

Component	Functional check
CU 382 control unit	The green LED is on and the text is visible on the display.
CU 382 control unit	The displayed system time is correct and the time is updated every second.
CU 382 control unit	No error messages or error symbols are displayed.
DID	All hoses and fittings are tight.
CU 382 control unit	The digital output of the flow detector is ok.
Outlet	Water is flowing out of the outlet hose. Water is visible within the hoses.
Flow cell	Water is flowing out of the outlet hose. Water is flowing out of the sampling point.
Hoses	The water flow is continuous, no air bubbles are visible in the hoses.
CU 382 control unit and sensors	No "NaN", no "- - - -" or "++++,++" is displayed. No Parameter name is flashing. All readings are plausible. Scroll through the displayed parameters with the [Up], [Down], [Left] and [Right] buttons.
CU 382 control unit and sensors	The chronological sequence looks plausible, with daily or seasonal fluctuation.
CU 382 control unit and sensors	All readings are within the specified and calibrated measuring range.
CU 382 control unit and sensors	There is a difference between the laboratory values and the readings of the sensor. • If the difference is significant, perform a local calibration. • Use a reliable and validated comparison method to verify the accuracy of the displayed readings.

11. Fault finding

See also the manual "Sensors V1" supplied with the sensor.

Fault	Cause	Remedy
1. No water flow	a) The inlet valve is closed.	Open the inlet valve completely.
	b) The inlet strainer is blocked	Clean the filter of the inlet strainer.
	c) The flow detector or the flow restrictor is blocked.	Clean the flow detector or the flow restrictor, or replace the complete component.
2. Water is running out of the sampling point.	a) The sampling point is not closed correctly.	Close the outlet fitting of the sampling point.
3. The power supply LED is off.	a) No power supply.	Connect to the power supply.
4. The displayed system time is not correct or is not updated every second.	a) The internal clock is not adjusted.	Adjust the clock in the "Setup" menu.
		Restart the CU 382 control unit.
5. Error messages or error symbols are displayed.	a) The sensor communicates an error.	In the menu "Parameter settings", select "Monitor" and check the sensor status. See also the manual "Sensors V1" for status information.
6. Hoses or connections are leaking.	a) The fittings are not tight.	Tighten the fittings carefully. Do not use force.
7. The flow detector shows no flow.	a) The inlet valve is closed.	Open the inlet valve.
	b) The inlet strainer is blocked.	Clean the inlet strainer.
8. Water is not visible in the hoses.	a) The inlet valve is closed.	Open the inlet valve.
	b) The inlet strainer is blocked.	Clean the inlet strainer.
9. Water is not visible in the hoses. Air comes out of the flow restrictor.	a) The pressure on the inlet is too high.	Reduce the pressure on the inlet.
10. The parameter readings are fluctuating.	a) The measuring electrode is contaminated or faulty.	Exchange the electrolyte or the membrane cap. See the manual "Sensors V1" for more information.
	b) The inlet strainer is blocked.	Clean the inlet strainer.
	c) Air bubbles are in the flow cell.	Reduce the pressure on the inlet.
11. The parameter readings are not plausible.	a) The measuring electrode is contaminated or faulty.	Exchange the electrolyte or the membrane cap. See the manual "Sensors V1" for more information.
	b) The inlet strainer is blocked.	Clean the inlet strainer.
	c) Air bubbles are in the flow cell.	Reduce the pressure on the inlet.
12. The readings are not within the specified and calibrated measuring range.	a) The sensor or the DID was powered-on very recently.	It takes up to an hour until a sensor provides correct values.
	b) The concentration of the measured substance is too high.	Change to a sensor with a wider measurement range.

12. Technical data

12.1 Hydraulic specifications of DID

Variants BF1 and BF3

Data	Unit	BF1	BF3
Number of connectable silicon sensors		1	3
General data	Minimum liquid temperature*	[°C]	0.1
	Maximum liquid temperature*	[°C]	45
	Minimum ambient temperature*	[°C]	0.1
	Maximum ambient temperature*	[°C]	45
Hydraulic data	Minimum flow rate	[l/h]	30
	Maximum flow rate (integrated flow restrictor)	[l/h]	60
	Maximum pressure at inlet (non fluctuating)	[bar]	3
	Minimum pressure at inlet (non fluctuating)	[bar]	0.1
	Maximum pressure at outlet (non fluctuating)	[bar]	0.5
Connection data	Internal thread of inlet and outlet	"	Rp1/2
	Size of inlet and outlet connection thread	-	G 5/8
	Inlet/outlet hose connections, metric, (ID/OD)	[mm]	4/6, 6/9, 6/12 or 9/12
	Inlet/outlet hose connections, inch, (IDxOD)	"	0.17 x 1/4, 1/4 x 3/8 or 3/8 x 1/2
	Sample water outlet, hose nipple	[mm]	4

* Depends also on the limits of the sensor and the CU 382 control unit. The lowest value defines the operating limit.

Variants TI1 and TI2

Data	Unit	TI1	TI2
Number of connectable silicon sensors		1	2
General data	Minimum liquid temperature*	[°C]	0.1
	Maximum liquid temperature*	[°C]	45
	Minimum ambient temperature*	[°C]	0.1
	Maximum ambient temperature*	[°C]	45
Hydraulic data	Maximum flow speed*	[m/s]	See manual "Sensors V1"
	Maximum pressure*	[bar]	10
Connection data	Prepared for pipe dimension, metric, (OD)	[mm]	50
	Prepared for pipe dimension, inch, (OD)	"	2

* Depends also on the limits of the sensor and the CU 382 control unit. The lowest value defines the operating limit.

12.2 CU 382 control unit

4-20 mA analog outputs

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of outputs		3	
Connection		Terminal block, stripped wire, AWG 28-12	
Galvanic isolation	[kV]	1 kV to system ground/earth	
Maximum load resistance	[Ohm]	500	
Minimum load resistance	[Ohm]	0	
Maximum output current	[mA]	20	
Resolution of reading		12-bit	
Response time	[ms]	< 1	
Drift over full temperature range		-50 µA to +10 µA at 10 mA	
Uncertainty	[µA]	< 30	

4-20 mA analog input

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of inputs		1	
Connection		Terminal block, stripped wire, AWG 28-12	
Input resistance	[Ohm]	100	
Galvanic isolation	[kV]	1 kV to system ground/earth	
Minimum input current	[mA]	0	
Maximum input current	[mA]	20	
Resolution of reading		12-bit	
Response time	[ms]	< 1	
Drift over full temperature range	[uA]	+30 µA to +0 µA at 10 mA	
Uncertainty	[uA]	< 30	

Digital output / Relay output

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of NO relays		2	
Number of NO/NC relays (error relays)		1	
Connection		Terminal block, stripped wire, AWG 28-12	
Maximal RMS current	[A]	< 6	
Maximal working voltage	[VAC]	< 240	
Maximal switchable load	[A]	< 6	
Maximal number of operating cycles	[VA]	< 600	
Required external protection		Depending on the load, additional snubber circuits may be required.	

Digital input - S4/S5 JP1:2 and JP5:6 set

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of inputs		2	
Connection		Terminal block, stripped wire, AWG 28-12	
Intended use		Digital input for external mechanical relay/switch	
Galvanic isolation	[kV]	none	
Maximum input frequency	[Hz]	100	
Low input voltage	[V]	< 7	
High input voltage	[V]	> 8	
Input resistance	[Ohm]	> 1000	

Digital input - S4/S5 JP1:2 and JP5:6 not set

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of inputs		2	
Connection		Terminal block, stripped wire, AWG 28-12	
Intended use		Digital input for external voltage output	
Galvanic isolation	[kV]	1 kV to system ground/earth	
Maximum input frequency	[Hz]	100	
High input	[Ohm]	> 2000	
Low input	[Ohm]	< 100	

Flow switch input

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of inputs		1	
Connection		Terminal block, stripped wire, AWG 28-12	
Intended use		For use with s::can flow detector	

Cleaning output

Description	Unit	D-320-GF1-230	D-320-GF3-230
Number of outputs		1	
Connection		Terminal block, stripped wire, AWG 28-12	
Intended use		For use with s::can cleaning valve B-44 or flow-cell autobrush F-446-2/ F-446-1	

Sensor

Description	Unit	D-320-GF1-230	D-320-GF3-230
Connection		1 x Buccaneer 400 Series 1 x Terminal block, stripped wire, AWG 28-12	
Intended use		For use with s::can sensors	
Maximum load	[W]	7 (constant), < 15 (peak)	

Control unit specifications

Description	Unit	D-320-GF1-230	D-320-GF3-230
Housing material		Polycarbonate, RAL 9005	
Material other		Polyamide, Polyurethane, EPDM	
Weight	[kg]	1.3	
Dimensions (WxHxL)	[mm]	213 x 185 x 118	
Mains connection		1.5 mm ² , 3-wire P/N/PE, fixed installation, M20 cable-gland entry	
Mains supply voltage	[VAC]	100-240/50-60 Hz	
Power consumption (typical)	[VA]	33, cos ϕ = 0.66	
Power consumption (maximum)	[A]	AC inrush current: < 30 AC current: < 0.15	
Terminal connections		4 x M12 cable-gland entry (one used by flow-switch) 1 x M16 cable-gland entry	
IP rating		IP65	
Memory	[MB]	512, industrial grade SLC	
Display	[mm]	128 x 64 graphical display, 70 x 40 viewing area, transreflective, white background	
Data transfer		USB host mode support for data transfer with USB stick, Supports FAT/ FAT32	
Firmware update		USB stick with FAT/FAT32 and firmware image.	
Integrated RTC		Uncertainty < 2 minutes/month at 25 °C	
Installation		A circuit breaker must be included in the installation. • The circuit breaker must be easily accessible. • The circuit breaker must be marked as the disconnecting device.	
Conformity - EMC		EN 61326-1:2013	
Conformity - safety		EN 61010-1:2010	
Conformity - RoHS2 (2011/65/EU)		EN 50581:2012	
Approvals		cTUVus, CE, CSA	

Operating conditions

Description	Unit	D-320-GF1-230	D-320-GF3-230
Operating temperature limits	[°C]	-20 to +45	
Humidity limits	[%]	5 to 90, non-condensing	
Maximum altitude	[m]	3000	
Manual cleaning		Cleaning with hand-warm water and mild detergents, such as dish- washing soap	

Storage

Description	Unit	D-320-GF1-230	D-320-GF3-230
Temperature limits	[°C]	-20 to +50	
Humidity limits	[%]	5 to 90, non-condensing	

12.3 Weights

DID Type	Without packaging	With packaging
	[kg]	[kg]
DID-1 BF1-FCL2	4.260	8.095
DID-1 BF1-CDI2	4.260	8.095
DID-1 BF1-PA2	4.260	8.095
DID-1 BF1-HP2	4.260	8.095
DID-3 BF3-FCL2/TCL2	4.720	8.775
DID-3 BF3-FCL2/PH	4.710	8.765
DID-3 BF3-FCL2/TCL2/PH	4.970	9.175
DID-3 BF3-FCL2/ORP	4.710	8.765
DID-3 BF3-FCL2/PH/ORP	4.960	9.165
DID-1 BF1-PH	4.250	8.085
DID-1 TI1-PH	2.075	5.875
DID-1 TI1-ORP	2.075	5.875
DID-3 TI2-ORP/PH	2.850	6.800
DID-3 BF3-ORP/PH	4.700	8.755
DID-1 BF1-CND	4.250	8.085
DID-1 TI1-CND	2.075	5.875

12.4 Dimensions

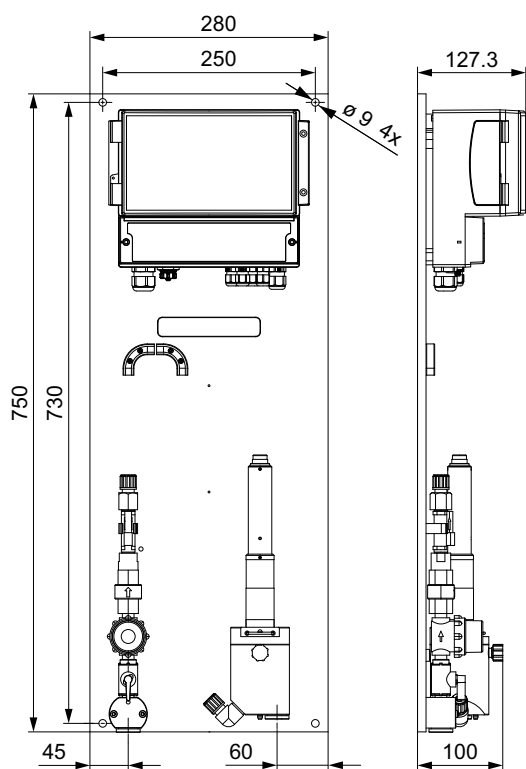


Fig. 10 Dimensions of DID with flow cell, BF1 variant for one sensor (in mm)

TM06 5852 02 16

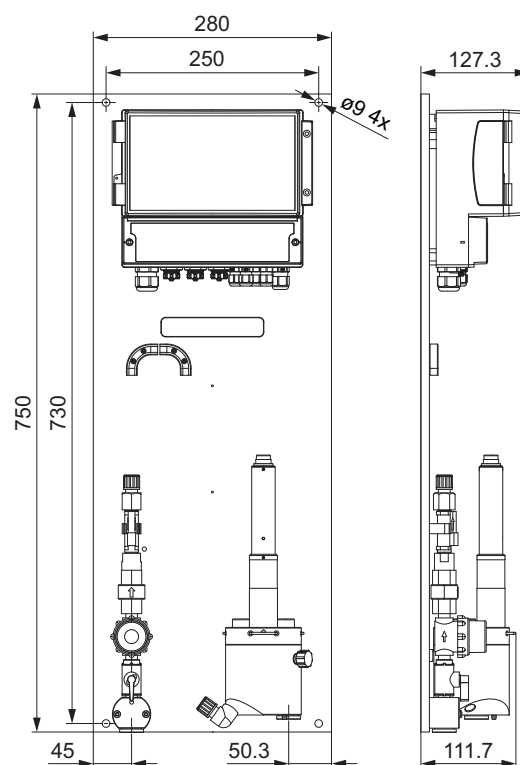


Fig. 11 Dimensions of DID with flow cell, BF3 variant for up to 3 sensors (in mm)

TM06 5853 02 16

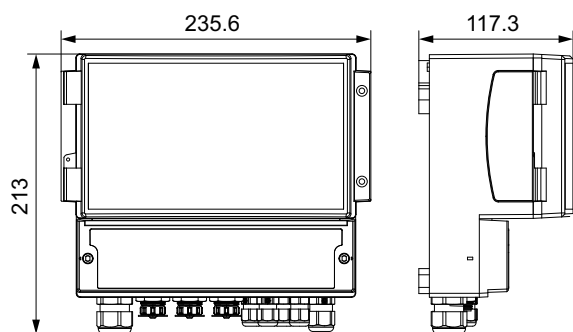


Fig. 12 Dimensions of CU 382 control unit, front and side (in mm)

TM06 5897 0216

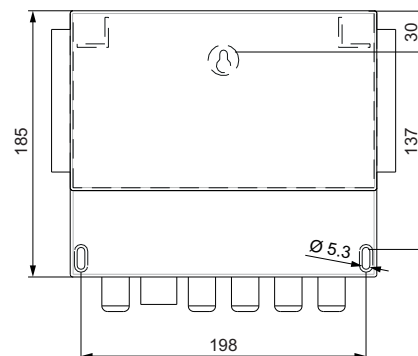


Fig. 13 Dimensions of CU 382 control unit for wall mounting, back (in mm)

TM06 6025 0416

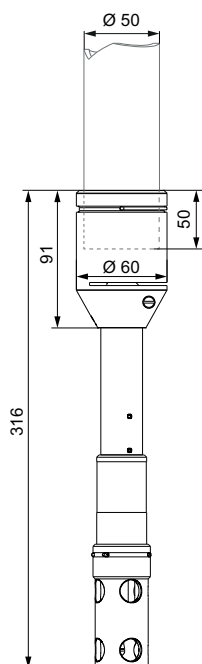


Fig. 14 Dimensions of sensor with holder (in mm), pipe with external diameter 50 mm not included

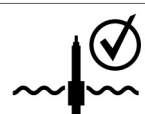
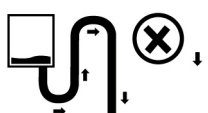
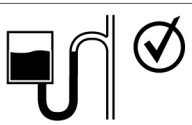
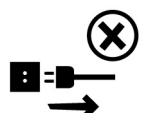
TM06 6039 0216

13. Disposal

This product or parts of it must be disposed of in an environmentally sound way. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

14. Appendix

14.1 Symbols on the product

Label	Description
	In case of problems call the Grundfos Service Hotline. Scanning the QR code forwards you to the Grundfos support site.
	For more information: Scanning the QR code forwards you to the Grundfos Product Center.
Min./Max. 0.1 °C – 45 °C 	Operating temperature: 0.1 °C to 45 °C
	Protect from freezing.
	Protect from direct sunlight.
	Read the installation and operating instructions before installing and operating the DID.
	Avoid drying-out of the sensors.
	The sensors must always be immersed in water.
	Avoid siphon effect.
	Use a vented outlet or a pressure-loading valve.
	Avoid power loss of the sensors that are equipped with electrolyte and membrane cap.

Label	Description
	Sensors equipped with electrolyte and membrane cap must remain powered.
Max. 0.5 bar 	The maximum pressure permitted inside the flow cell is 0.5 bar. The inlet pressure is limited by the flow restrictor. The operator must make sure that no counterpressure is on the outlet.
~10 ml 3-4 	Before taking a sample for calibration, flush the sampling point 3-4 times with approximately 10 ml each time.
 Max. 3 bar Min. 0.5 l/min	Maximum inlet pressure: 3 bar Minimum flow rate: 0.5 litres per minute

Declaration of conformity

GB: EC declaration of conformity

We, s::can Messtechnik GmbH, hereby declare that the product listed below, to which this Declaration of Conformity relates, is in conformity with Directives, Standards and other Normative Documents as listed.

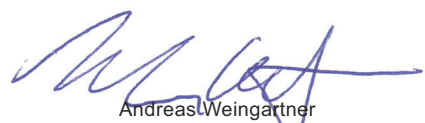
Type of product: Measurement & Process Control

Name of product: DID

Model number: D-320-GF1-230, D-320-GF3-230

- Low Voltage Directive (2014/35/EU)
Standard used: EN 61010-1:2010
- RoHS2 (2011/65/EU)
Standard used: EN 50581:2012
- EMC Directive (2014/30/EU).
Standard used: EN 61326-1:2013

Vienna, 20th March 2017



Andreas Weingartner
(Director s::can Messtechnik GmbH, Vienna)

Person authorised to compile technical file and
empowered to sign the EC declaration of conformity.

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